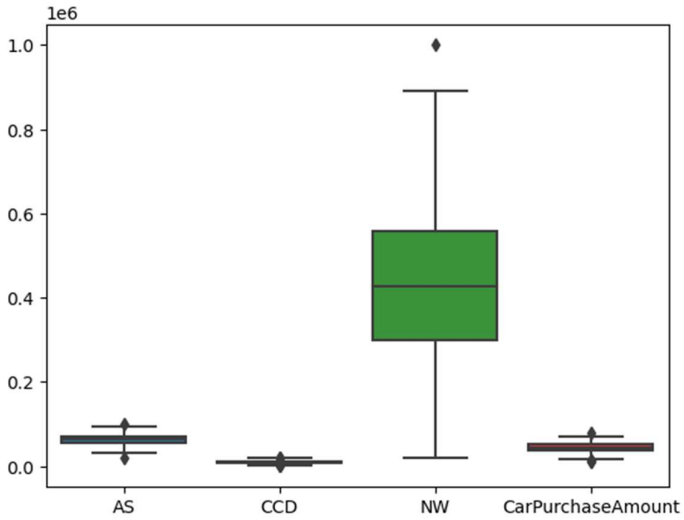
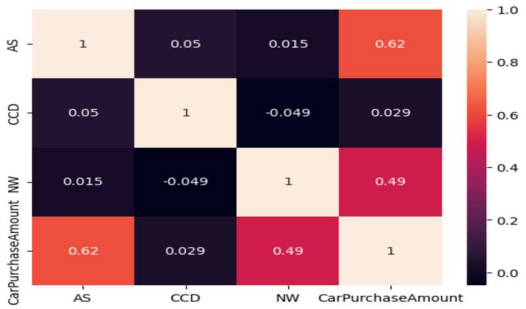
PYTHON FINAL SUBMISSION

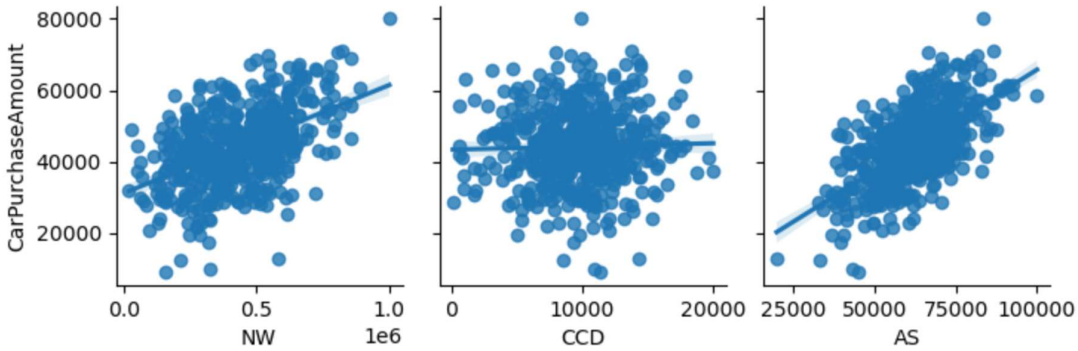
Dataset: [Car Purchasing Data]



In the first step, we needed to check if there are any outliers in the data. There were outliers in all columns and therefore we decided they needed to be treated.



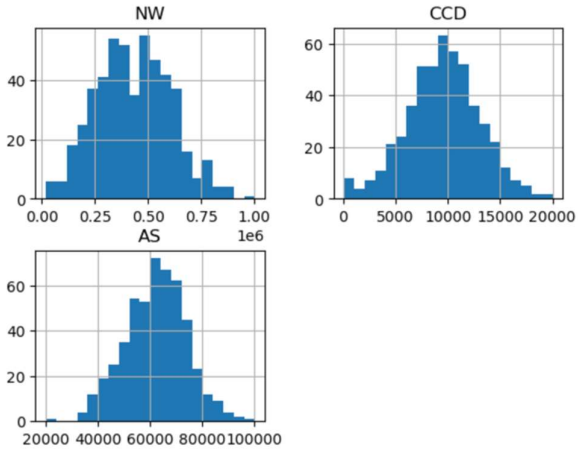
The next step was to see if there is any correlation among the dependent variables. The one pair that had a high correlation was Annual Salary(AS) and Car Purchase Amount with a coefficient of 0.62. Net worth(NW) and car purchase amount also do have a significant correlation.



The above scatter plots tell us that Annual Salary (AS) has a lot of points that fall on the best-fit line whereas in both net worth and credit card debt do have a few lines that are far away from the best-fit line.

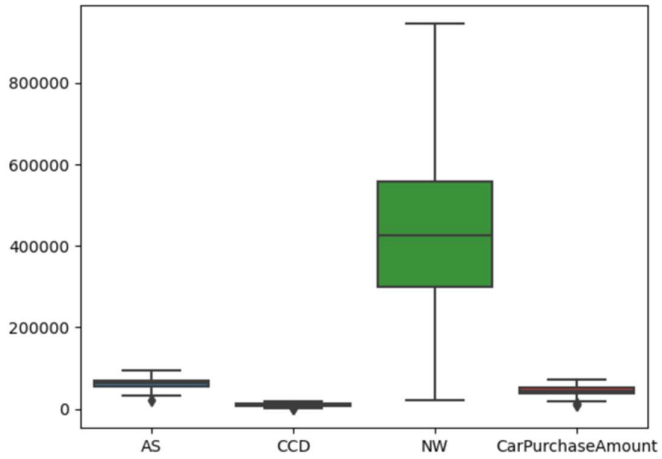
	features	VIF Factor
0	NW	6.10
1	CCD	7.23
2	AS	10.85

Both Net worth and Credit card debt lie in the acceptable region for VIF but Annual Salary is slightly above the range but not by much so keeping in mind the above scatter plot we will keep this variable in the equation.



From the above charts, we can see that all data points are normally distributed in each of the three columns.

After this, we treated the outliers to get better accuracy for the model. From the below diagram, we can see that the outliers are now within the acceptable range.



OLS Regression Results

Dep. Variable:	CarPurchaseAmount	**R-squared:**	0.607
Model:	OLS	**Adj. R-squared:**	0.605
Method:	Least Squares	**F-statistic:**	267.8
Date:	Sat, 15 Apr 2023	**Prob (F-statistic):**	4.54e-71
Time:	14:03:28	**Log-Likelihood:**	-3577.7
No. Observations:	350	**AIC:**	7161.
Df Residuals:	347	**BIC:**	7173.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	-885.0247	2077.655	-0.426	0.670	-4971.407	3201.357
AS	0.5289	0.030	17.386	0.000	0.469	0.589
NW	0.0292	0.002	14.531	0.000	0.025	0.033

Omnibus:	0.132	**Durbin-Watson:**	2.008
Prob(Omnibus):	0.936	**Jarque-Bera (JB):**	0.166
Skew:	0.046	**Prob(JB):**	0.920
Kurtosis:	2.947	**Cond. No.**	2.72e+06

The above table gives us the following facts:

- The dependent variable is "Car Purchase Amount", and there are three independent variables: "Annual Salary(AS)", "(Credit Card Debt (CCD)", and "Net Worth (NW)"

- The model has an R-squared value of 0.607, which means that approximately 60.7% of the variation in the dependent variable is explained by the independent variables. Also, the adjusted R squared is 0.605 and we can see the difference between both R squared and adjusted R squared is minimal which means the model has better accuracy.

- The F-statistic is 178.4 which is the tabulated F value and the Prob F-value is 6.55e-70 which is the calculated F value. The above figures tell us that the overall model is statistically significant and as the gap is high the accuracy of the model is higher.

- The coefficients of "Annual Salary(AS)", "Credit Card Debt (CCD)", and "Net Worth (NW)" are 0.5283, 0.0760, and 0.0292, respectively.

- The p-value for "AS" and “NW” is (0.000) which makes these variables statistically significant, while the p-value for "CCD" is **0.479** which makes it statistically insignificant.

- The Omnibus and Jarque-Bera tests are measures of normality, and their associated p-values indicate that the residuals are normally distributed.

- The Durbin-Watson tests the model for correlation between the residuals. This value is 2.004 which suggests that there is no autocorrelation between the errors.

- The Skewness value is **0.0408** which means the data is positively skewed. In a positive skew, the tail of a distribution curve is longer on the right side. This means the outliers of the distribution curve are further out towards the right and closer to the mean. The value is within the acceptable range.

The AIC and BIC are measures of model fit, with lower values indicating a better fit. Overall, the regression model suggests that the variable "AS" and “NW” are statistically significant predictors of "Car Purchase Amount", while "CCD" does not appear to have a statistically significant relationship with the "Car Purchase Amount". However, it's important to note that additional analysis (such as checking for multicollinearity, outliers, or influential observations) should be conducted to further validate the model.